
ANNAMALAI RAGHUPATHY

AI Engineer | Agentic AI Developer | Azure Data Engineer

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Location: Coimbatore, India

PROFESSIONAL SUMMARY

AI-focused Azure Data Engineer with 4+ years of experience building scalable data pipelines and transitioning into AI/ML engineering. Skilled in developing agentic AI systems, LLM-powered applications, and intelligent automation workflows using LangChain, LangGraph, and Azure Databricks. Proven experience in production support, pipeline monitoring, and designing AI-driven solutions for real-world business problems. Strong foundation in Python, machine learning, and cloud-based AI deployment.

TECHNICAL SKILLS

AI / ML:

Agentic AI, LangChain, LangGraph, CrewAI, MLflow, YOLO, LLM Fine-tuning (DeepSeek), Computer Vision

Programming:

Python, SQL, PySpark, Scala

Cloud & Data Platforms:

Azure Databricks, Azure Data Factory (ADF), Azure Synapse, Delta Lake, Unity Catalog

DevOps & Tools:

Azure DevOps, CI/CD Pipelines, Docker, Git

APIs & Frameworks:

FastAPI, Flask, REST APIs, Web Scraping, SERP API

Visualization:

Power BI (DAX, KPIs, dashboards)

PROFESSIONAL EXPERIENCE

Azure Data Engineer

Unilever & Meijer Projects | Jan 2019 – Present

- Designed and developed scalable ETL pipelines using Azure Data Factory and Databricks.
- Built PySpark-based data processing workflows for large-scale enterprise systems.
- Automated CI/CD pipelines using Azure DevOps, improving deployment speed and reliability.
- Implemented data governance and security using Unity Catalog and Azure Synapse.
- Migrated legacy data systems to modern cloud-based architectures.

Production Support (Meijer Project):

- Managed and resolved production issues, ensuring high availability of critical data pipelines.
 - Performed root cause analysis and implemented fixes to prevent recurring failures.
 - Monitored pipeline performance and ensured SLA compliance for business-critical workflows.
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AI PROJECTS & INNOVATIONS

Agentic AI Pipeline Monitoring System (Azure Databricks)

- Built a Proof of Concept for pipeline monitoring using agentic AI.
- Enabled automated failure detection, anomaly identification, and intelligent alerting.
- Designed an LLM-based solution recommendation system to auto-suggest fixes for pipeline issues.

AI Agent Development (Databricks + LangChain)

- Developed first AI agent in Azure Databricks for workflow automation and orchestration.
- Implemented multi-agent collaboration using LangGraph and CrewAI.

Freshness Detection System (Hackathon Project)

- Secured **2nd place in Agentic AI Hackathon**.
- Built a hybrid AI system for **meat and banana freshness detection** using YOLO and ML models.
- Integrated computer vision with agentic AI for intelligent decision workflows.

LLM & Automation Projects

- Built FastAPI-based microservices for AI agent deployment.
- Developed web scraping agents integrated with SERP APIs.

- Experimented with fine-tuning lightweight LLMs (DeepSeek).
 - Integrated AI agents with Azure DevOps and Databricks pipelines.
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EDUCATION

Bachelor of Mechatronics Engineering
Kumaraguru College of Technology | Coimbatore | 2016 – 2020

CERTIFICATIONS

- Microsoft Azure Fundamentals (DP-900)
 - Anthropic - Claude Code in Action
 - Google Cloud Skills - Generative AI Basics and Essentials
 - Microsoft - Azure AI Computer Vision
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AWARDS & ACHIEVEMENTS

- Top Performer Award – Meijer Project
 - 2nd Place – Agentic AI Hackathon
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